

Lecture 4:

Sequence Methods

Part of Speech Tagging and
Hidden Markov Models

Part of Speech (POS) Tagging

INPUT:

Profits soared at Boeing Co., easily topping forecasts on Wall Street, as their CEO Alan Mulally announced first quarter results.

OUTPUT:

Profits/**N** soared/**V** at/**P** Boeing/**N** Co./**N** ,/, easily/**ADV** topping/**V**
forecasts/**N** on/**P** Wall/**N** Street/**N** ,/, as/**P** their/**POSS** CEO/**N**
Alan/**N** Mulally/**N** announced/**V** first/**ADJ** quarter/**N** results/**N** ./.

N = Noun

V = Verb

P = Preposition

Adv = Adverb

Adj = Adjective

...

Part of Speech (POS) Tagging

Training set:

1 Pierre/**NNP** Vinken/**NNP** ,/, 61/**CD** years/**NNS** old/**JJ** ,/, will/**MD** join/**VB** the/**DT** board/**NN** as/**IN** a/**DT** nonexecutive/**JJ** director/**NN** Nov./**NNP** 29/**CD** ./.

2 Mr./**NNP** Vinken/**NNP** is/**VBZ** chairman/**NN** of/**IN** Elsevier/**NNP** N.V./**NNP** ,/, the/**DT** Dutch/**NNP** publishing/**VBG** group/**NN** ./.

3 Rudolph/**NNP** Agnew/**NNP** ,/, 55/**CD** years/**NNS** old/**JJ** and/**CC** chairman/**NN** of/**IN** Consolidated/**NNP** Gold/**NNP** Fields/**NNP** PLC/**NNP** ,/, was/**VBD** named/**VBN** a/**DT** nonexecutive/**JJ** director/**NN** of/**IN** this/**DT** British/**JJ** industrial/**JJ** conglomerate/**NN** ./.

...

38,219 It/**PRP** is/**VBZ** also/**RB** pulling/**VBG** 20/**CD** people/**NNS** out/**IN** of/**IN** Puerto/**NNP** Rico/**NNP** ,/, who/**WP** were/**VBD** helping/**VBG** Hurricane/**NNP** Hugo/**NNP** victims/**NNS** ,/, and/**CC** sending/**VBG** them/**PRP** to/**TO** San/**NNP** Francisco/**NNP** instead/**RB** ./.

- From the training set, induce a function/algorithm that maps new sentences to their tag sequences.

Tagging – the Supervised Setup

Training set:

1 Pierre/NNP Vinken/NNP ,/, 61/CD years/NNS old/JJ ,/, will/MD join/VB the/DT board/NN as/IN a/DT nonexecutive/JJ director/NN Nov./NNP 29/CD ./.

2 Mr./NNP Vinken/NNP is/VBZ chairman/NN of/IN Elsevier/NNP N.V./NNP ,/, the/DT Dutch/NNP publishing/VBG group/NN ./.

3 Rudolph/NNP Agnew/NNP ,/, 55/CD years/NNS old/JJ and/CC chairman/NN of/IN Consolidated/NNP Gold/NNP Fields/NNP PLC/NNP ,/, was/VBD named/VBN a/DT nonexecutive/JJ director/NN of/IN this/DT British/JJ industrial/JJ conglomerate/NN ./.

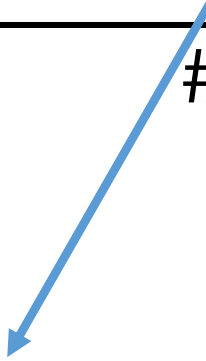
...

38,219 It/PRP is/VBZ also/RB pulling/VBG 20/CD people/NNS out/IN of/IN Puerto/NNP Rico/NNP ,/, who/WP were/VBD helping/VBG Hurricane/NNP Hugo/NNP victims/NNS ,/, and/CC sending/VBG them/PRP to/TO San/NNP Francisco/NNP instead/RB ./.

Evaluation:

accuracy

$$= \frac{\# \text{ test set words with correct tag}}{\# \text{ test set words}}$$



Word tokens as opposed to *word types*. That is of the *word type* dog appears 5 times in the test set, it will be counted 5 times by the accuracy measure

Named Entity Recognition (NER)

INPUT: Profits soared at Boeing Co., easily topping forecasts on Wall Street, as their CEO Alan Mulally announced first quarter results.

OUTPUT: Profits soared at [Company Boeing Co.], easily topping forecasts on [Location Wall Street], as their CEO [Person Alan Mulally] announced first quarter results.

NER as a Sequence Labeling Task

INPUT:

Profits soared at Boeing Co., easily topping forecasts on Wall Street, as their CEO Alan Mulally announced first quarter results.

OUTPUT:

Profits/NA soared/NA at/NA Boeing/SC Co./CC ,/NA easily/NA
topping/NA forecasts/NA on/NA Wall/SL Street/CL ,/NA as/NA
their/NA CEO/NA Alan/SP Mulally/CP announced/NA first/NA
quarter/NA results/NA ./NA

NA = No entity
SC = Start Company
CC = Continue Company
SL = Start Location
CL = Continue Location
...

Task Definition: Part of Speech Tags

- Parts of Speech are word categories that have similar morphological or syntactic behavior
- For instance, we can define an adjective as a word which appears as a modifier in a Noun Phrase, as a predicate in a copula clause or in a comparative construction
 - The _____ dog (*black/red/hungry/tall*)
 - The dog is _____ (*black/red/hungry/tall*)
 - X / X+er / X + est (*black/red/hungry/tall*)
- Circular definition! syntactic environments are defined in terms of grammatical categories

Part of Speech Tags

- Wordforms often have more than one possible POS: *back*
 - The *back* door = *Adj*
 - On my *back* = *Noun*
 - Win the voters *back* = *Adverb*
 - Promised to *back* the bill = *Verb*
- The POS tagging problem is to determine the (single) POS tag for a particular word token (instance)

Sources of information

- What are the main sources of information for POS tagging?
 1. Knowledge of word probabilities
 - *man* is rarely used as a verb...
 2. Knowledge of neighboring words

Bill	saw	that	man	yesterday
name	verb (past)	det.	noun	adverb
<i>verb</i>	<i>verb</i>	<i>det.</i>	<i>verb</i>	<i>adverb</i>
<i>verb</i>	<i>noun</i>	<i>conj.</i>	<i>verb</i>	<i>adverb</i>

Word-level Classification

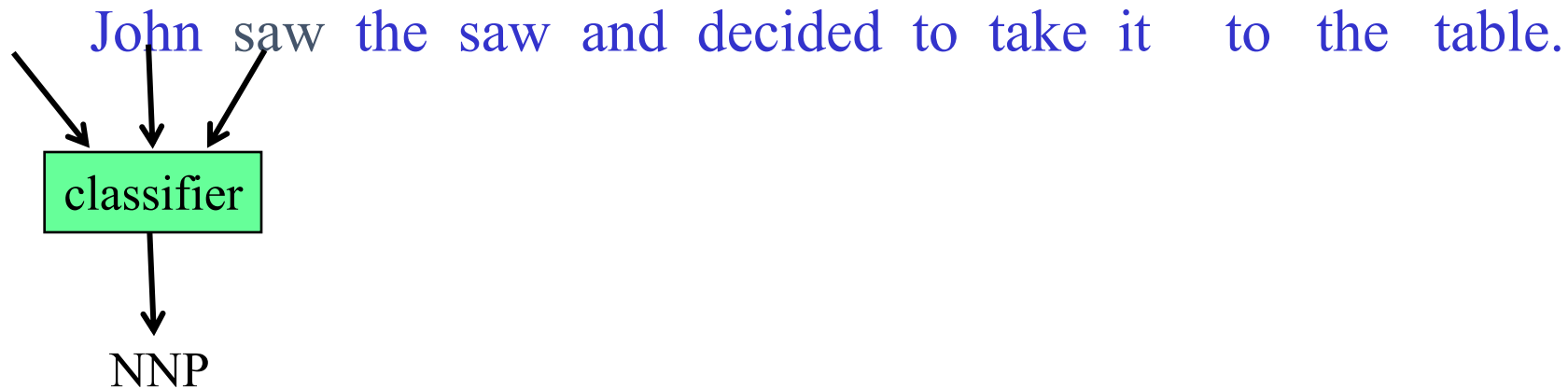
- We can do pretty well by classifying each word on its own
- Features:
 - Lowercase / uppercase
 - Prefixes / suffixes ('-ed' → verb, '-ly' → adverb, 'un-' → adjective)
 - Non-letter characters (periods → acronyms, only numbers → quantifier)

Bill	saw	that	man	yesterday
name	verb (past)	det.	noun	adverb
<i>verb</i>	<i>verb</i>	<i>det.</i>	<i>verb</i>	<i>adverb</i>
<i>verb</i>	<i>noun</i>	<i>conj.</i>	<i>verb</i>	<i>adverb</i>

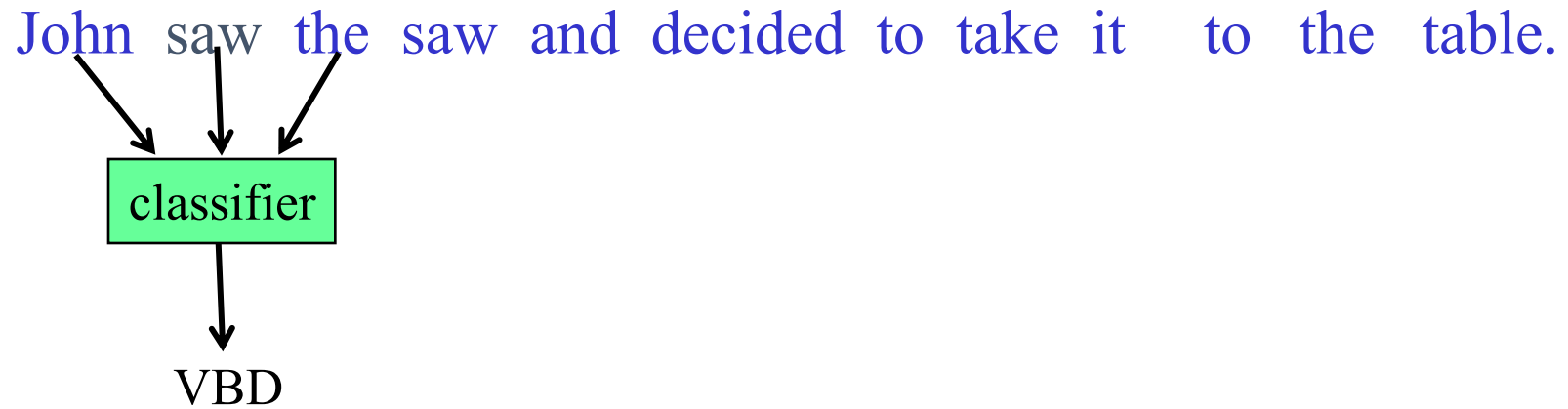
Not necessarily a bad option.
Think of: “find Mary before
coming back”

Sequence Labeling as Classification

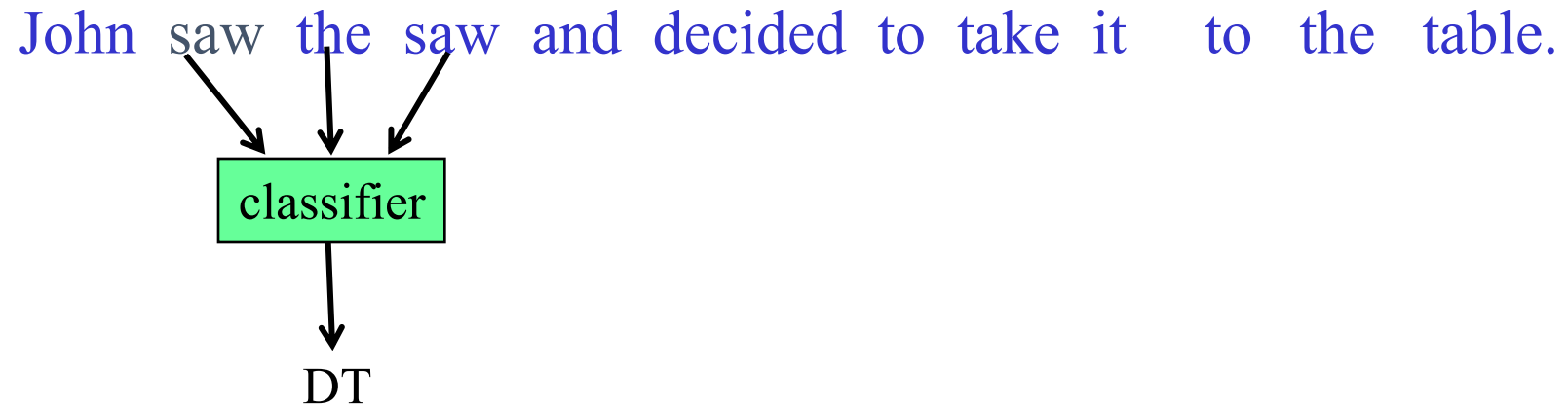
- **Option 1:** classify each token independently but use as input features, information about the surrounding tokens (sliding window)



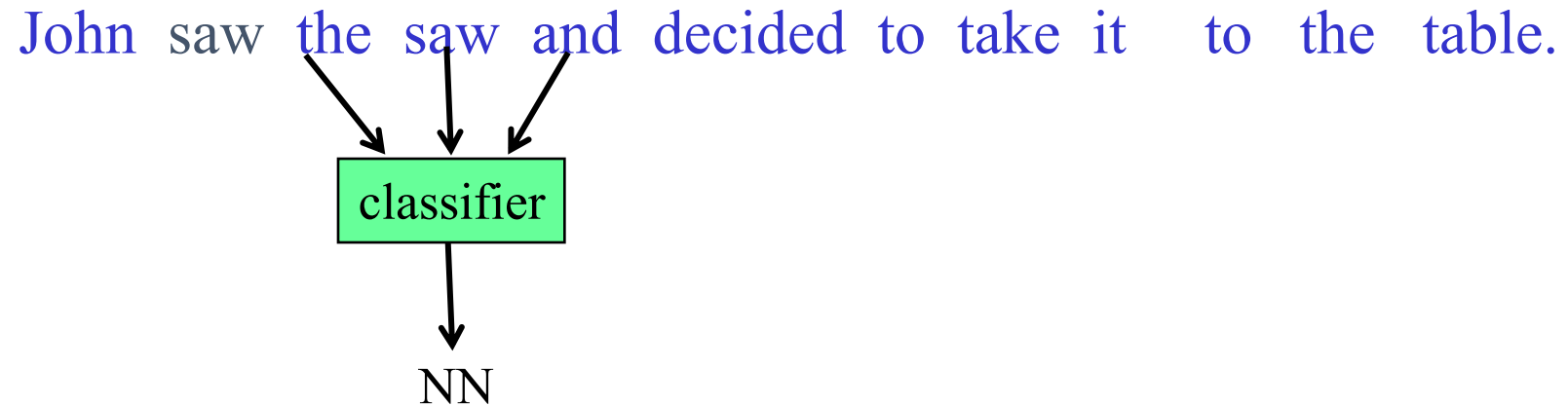
Sequence Labeling as Classification



Sequence Labeling as Classification

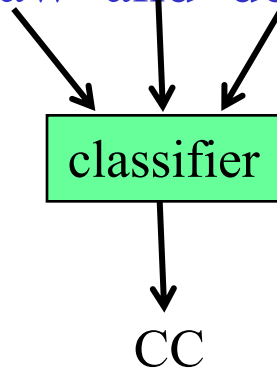


Sequence Labeling as Classification

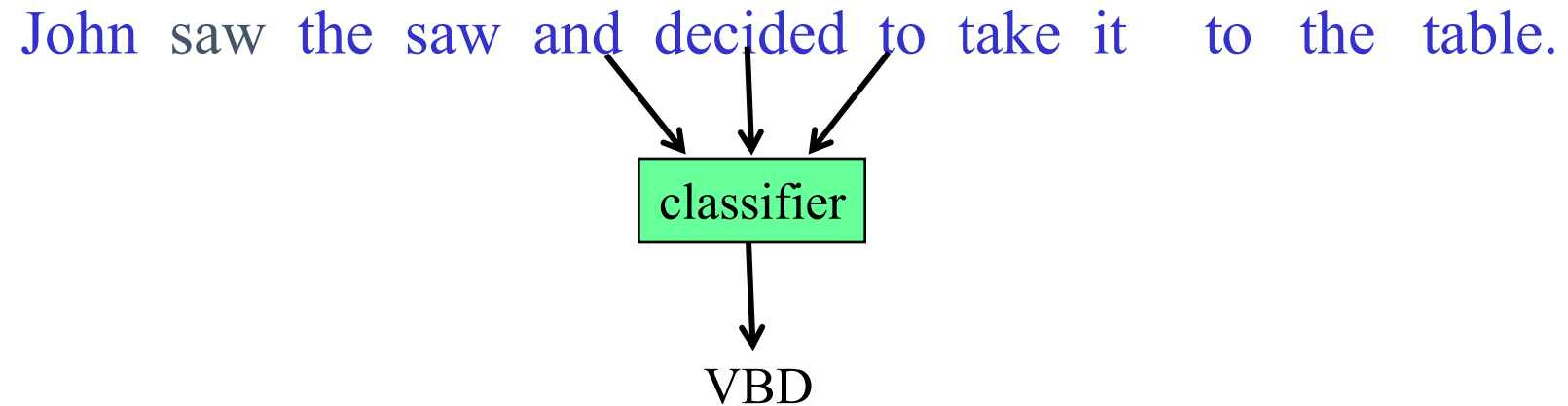


Sequence Labeling as Classification

John saw the saw and decided to take it to the table.

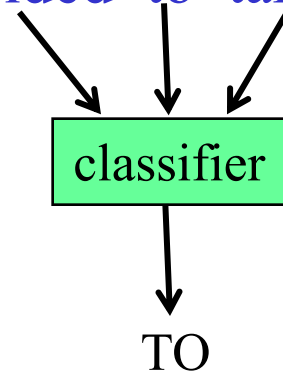


Sequence Labeling as Classification



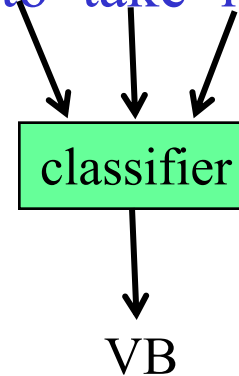
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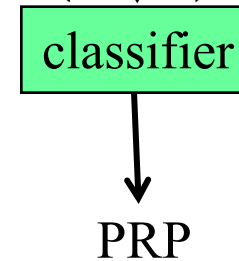
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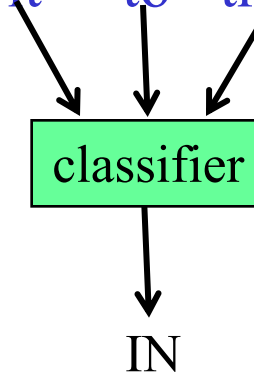
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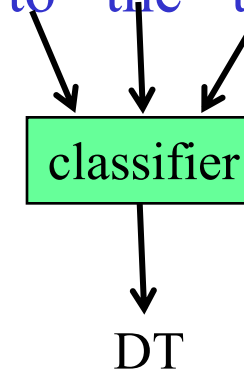
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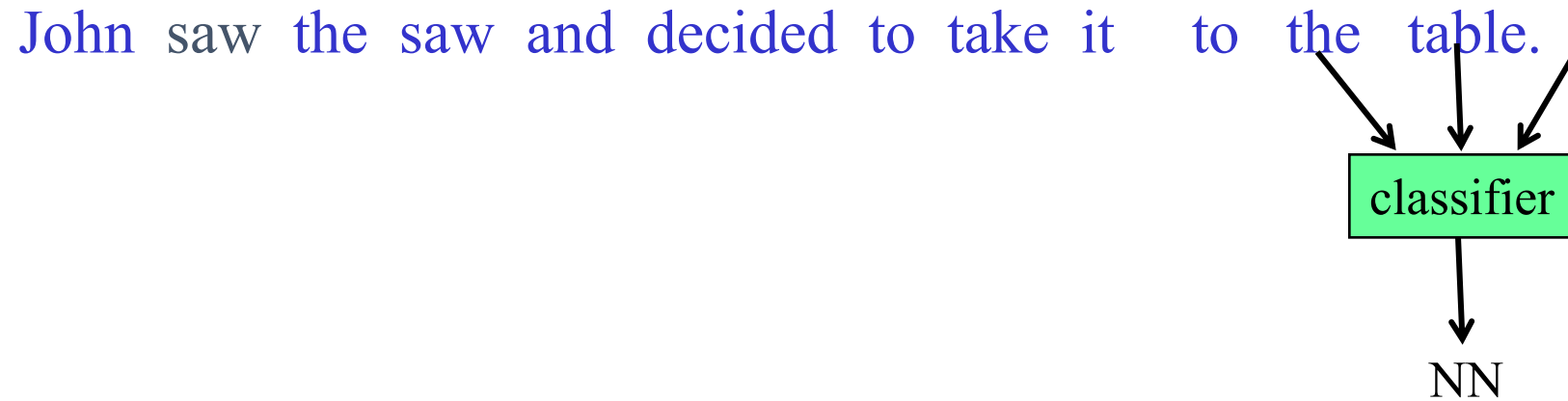


Sequence Labeling as Classification

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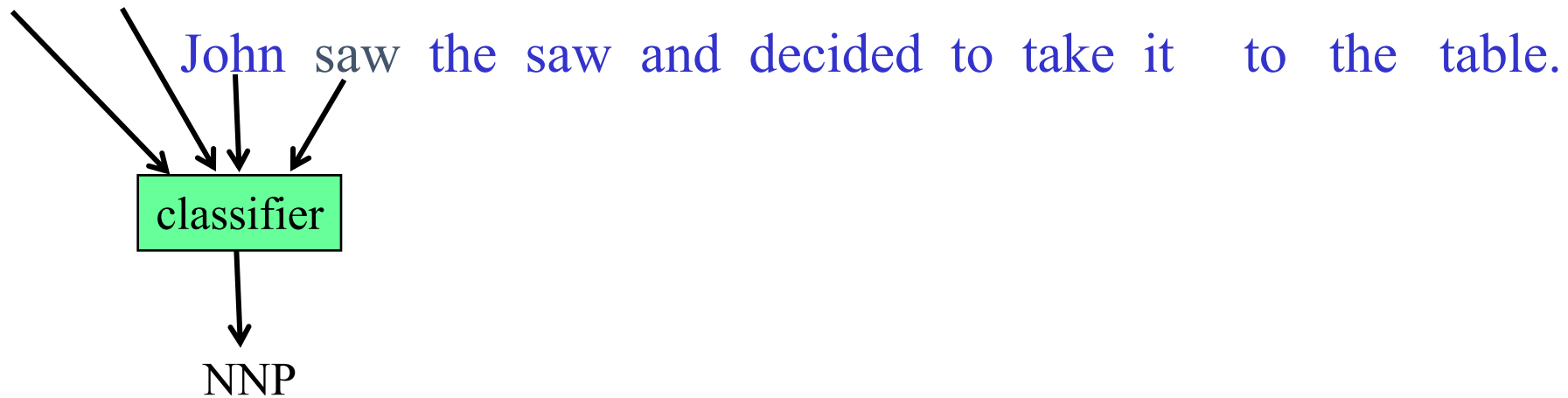
Sequence Labeling as Classification



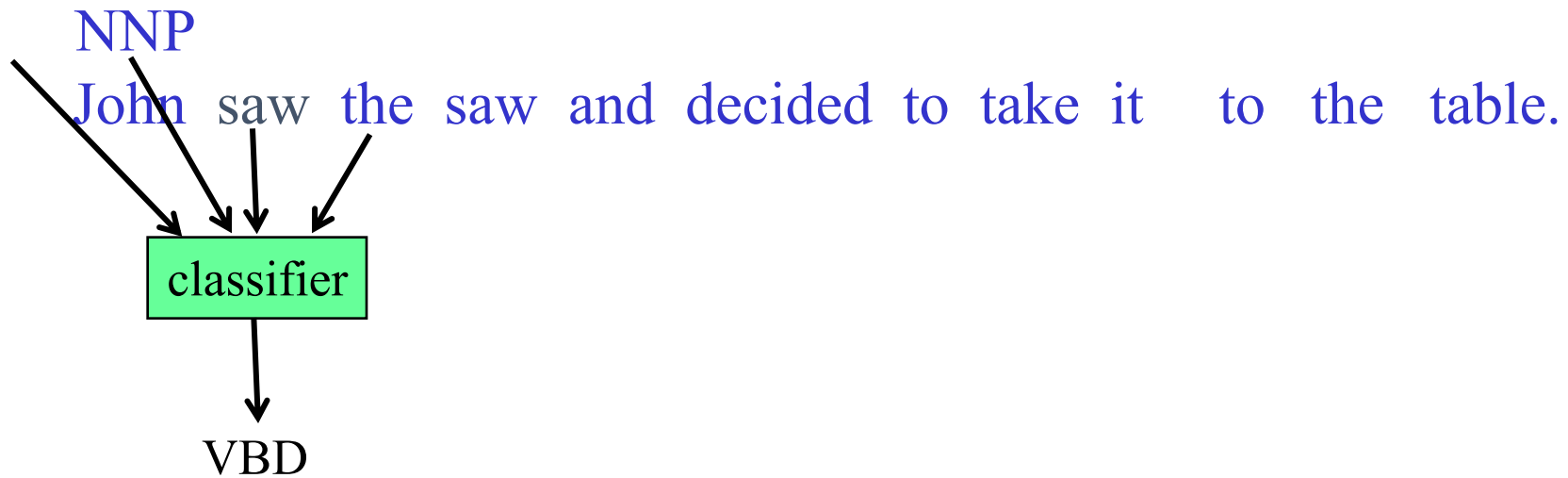
Sequence Labeling as Classification using Outputs as Inputs

- Better input features are usually the **categories** of the surrounding tokens, but these are not available yet
- Can use category of either the preceding or succeeding tokens by going forward or backward and using previous output

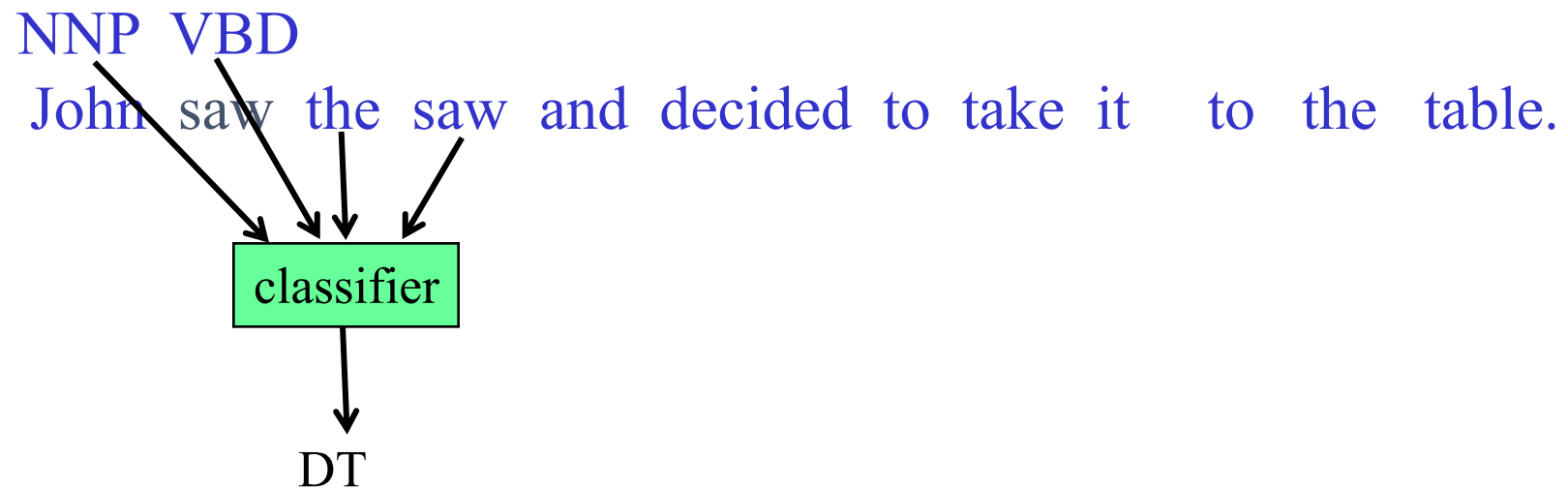
Forward Classification



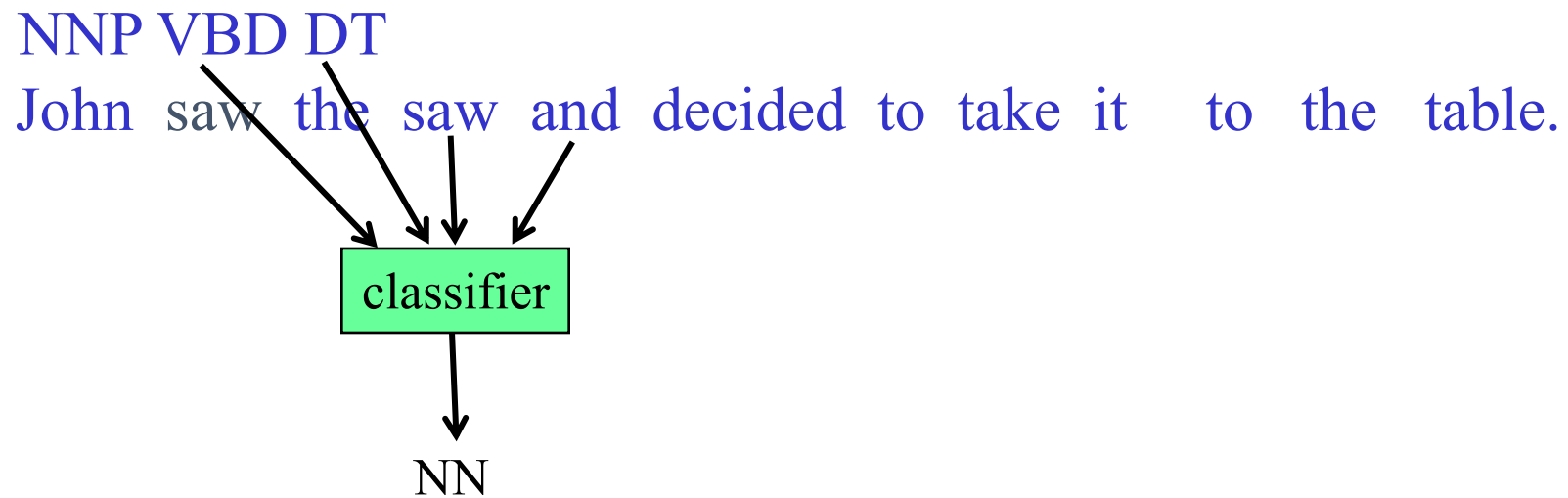
Forward Classification



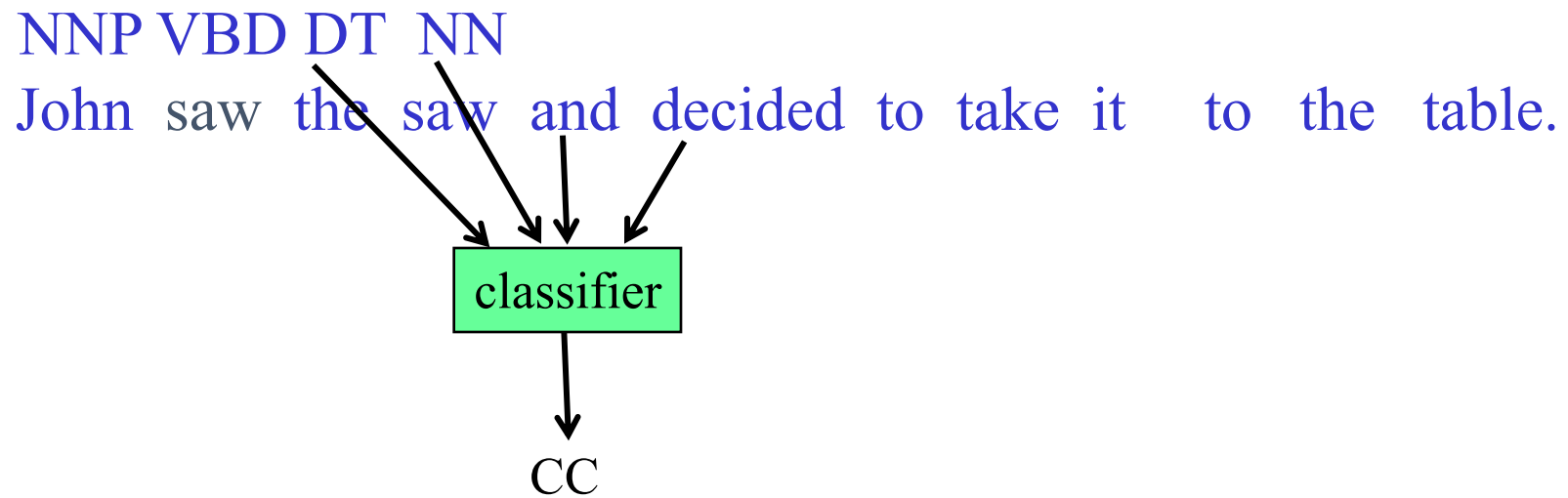
Forward Classification



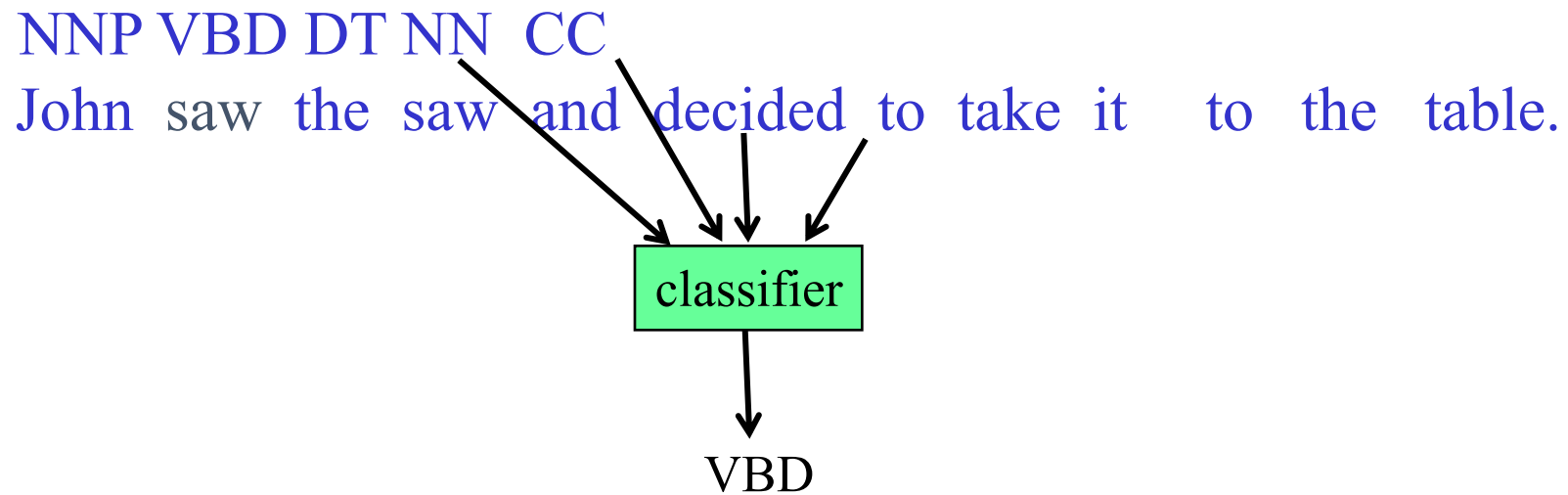
Forward Classification



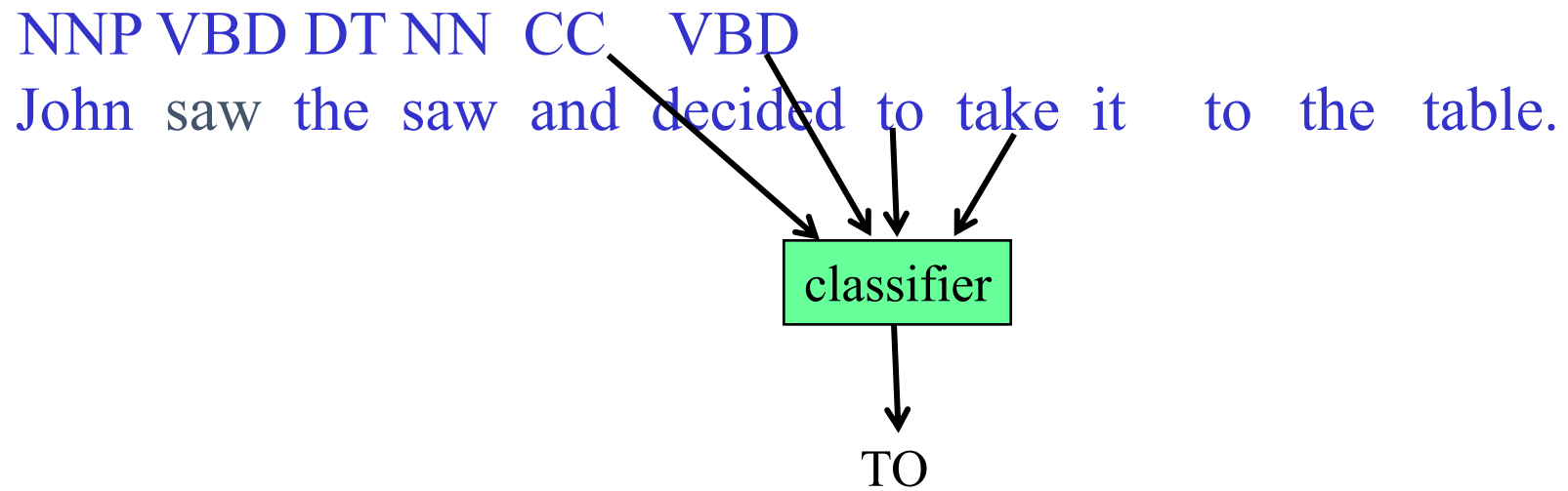
Forward Classification



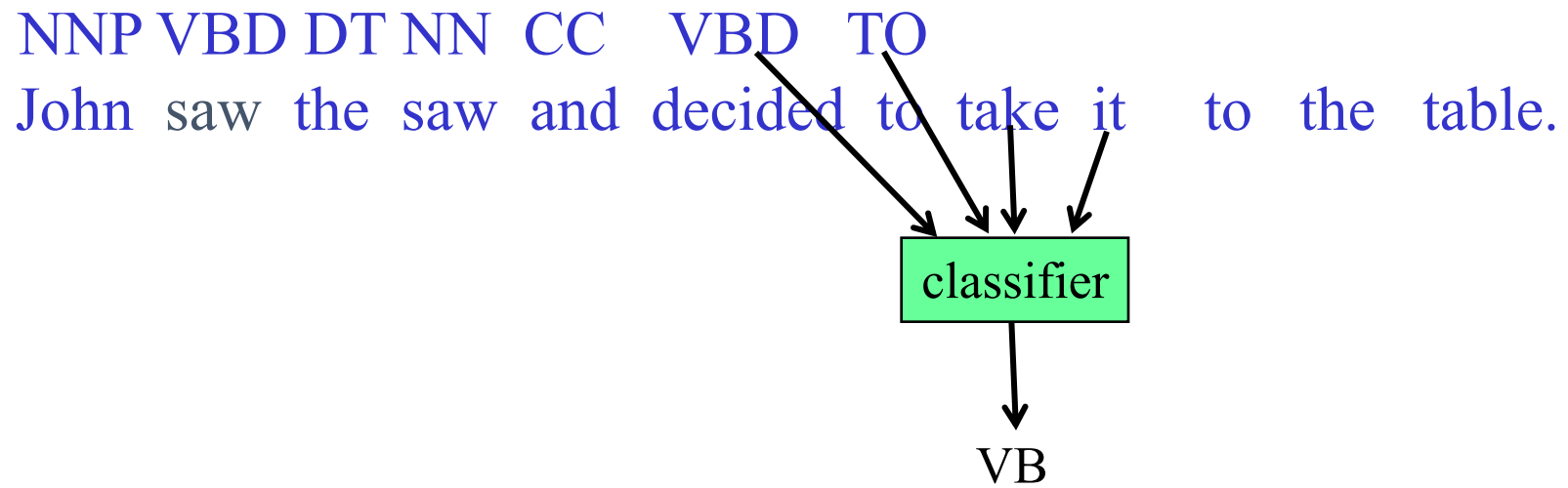
Forward Classification



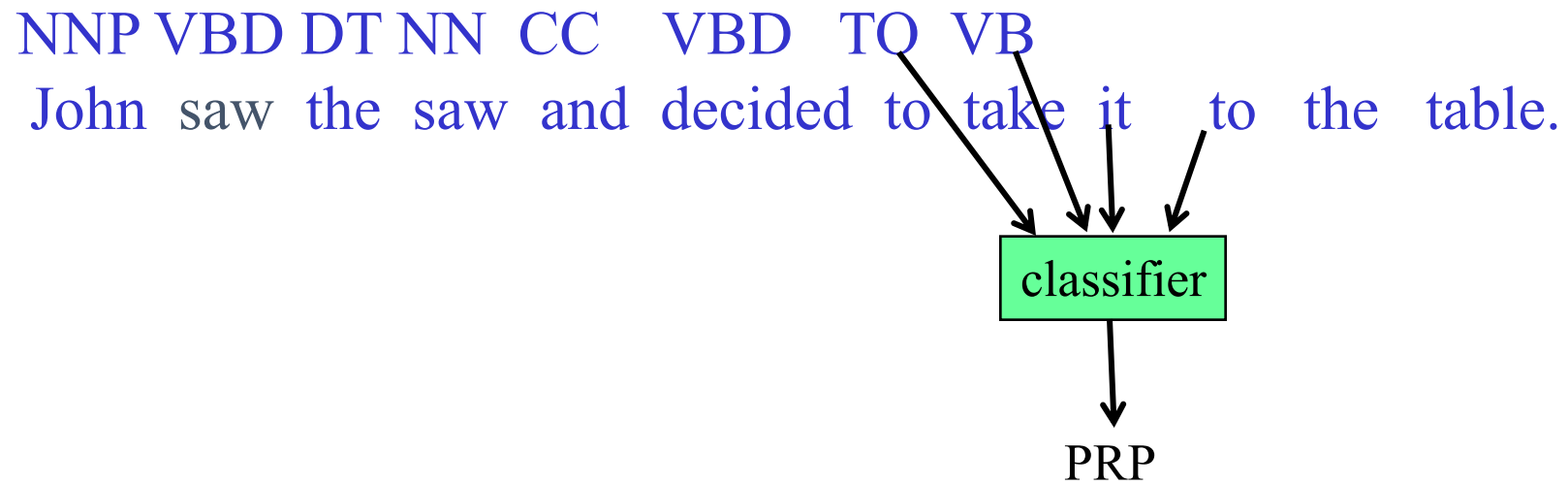
Forward Classification



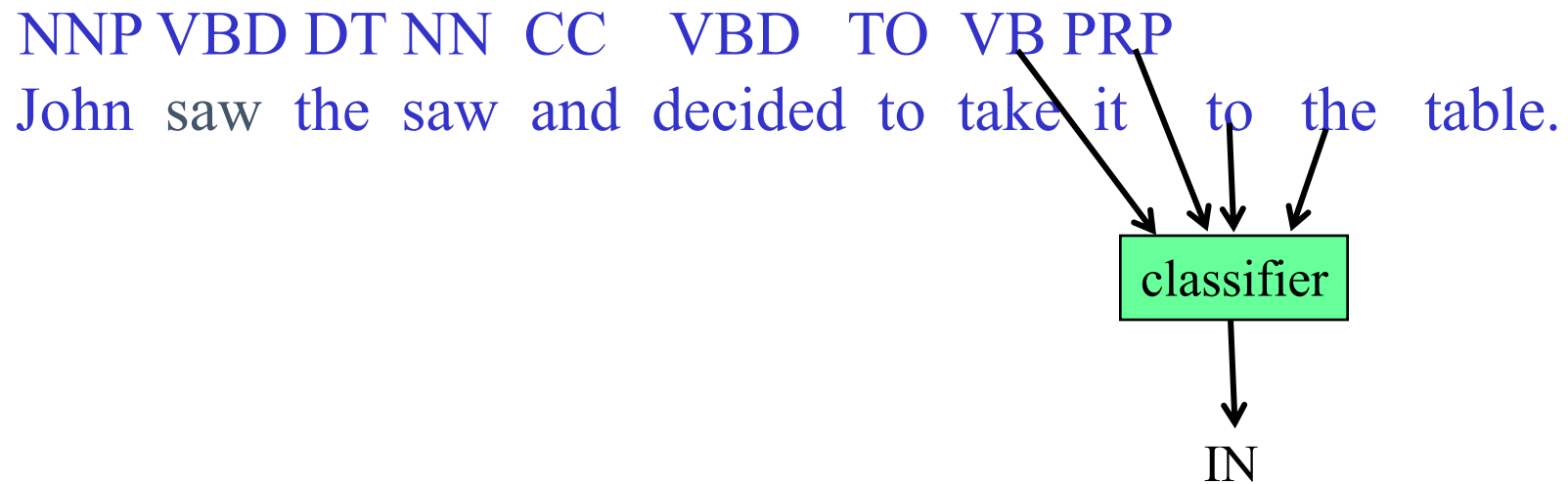
Forward Classification



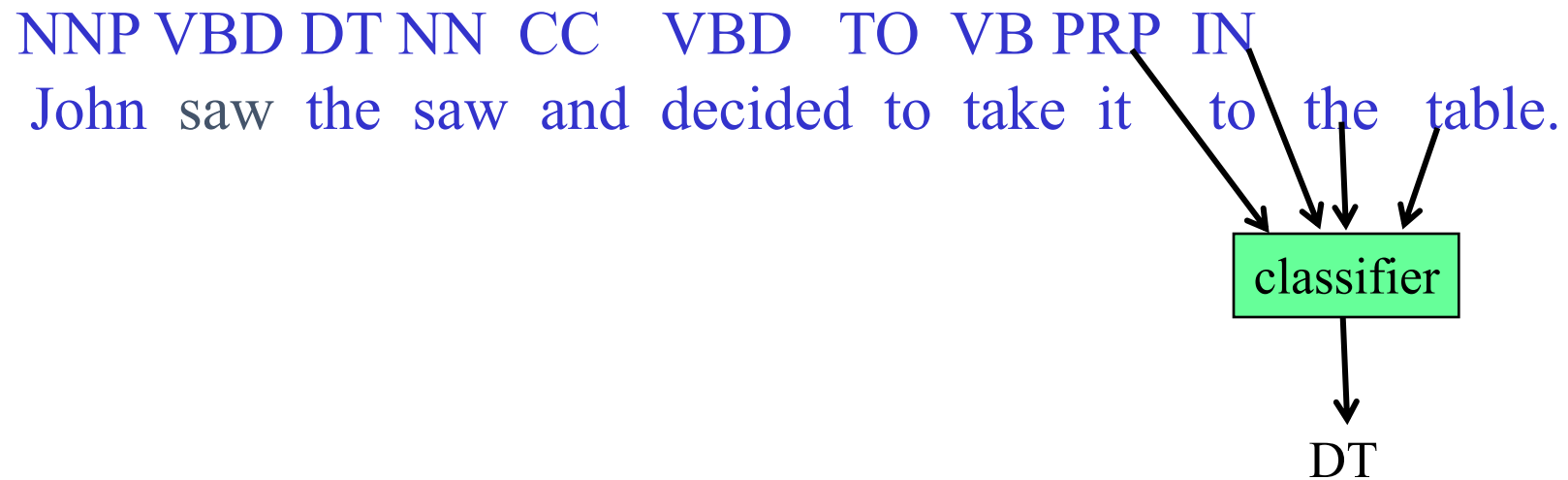
Forward Classification



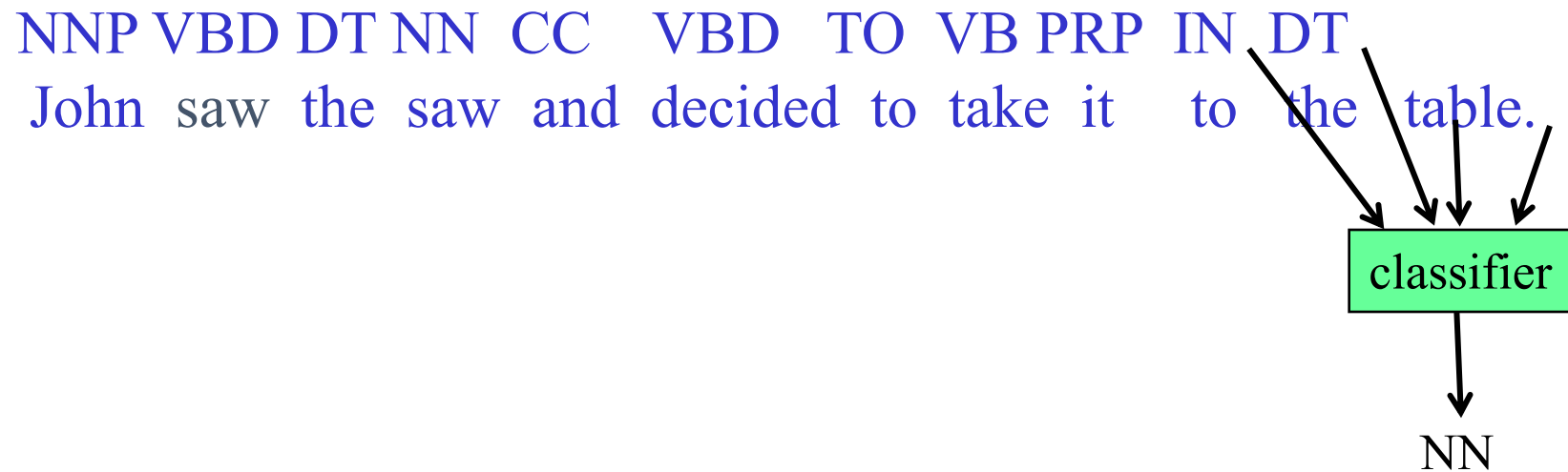
Forward Classification



Forward Classification

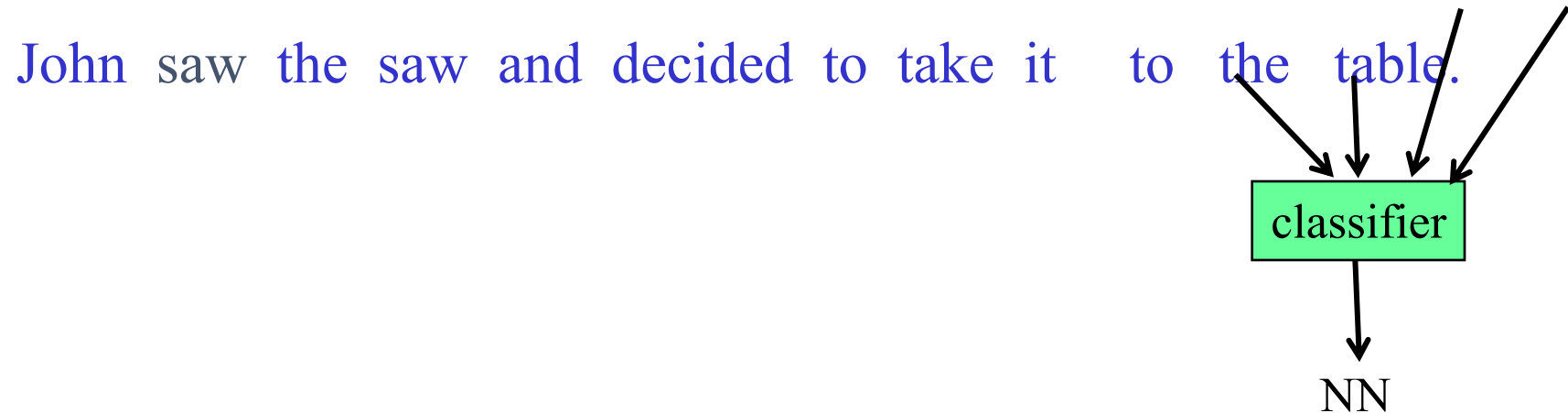


Forward Classification

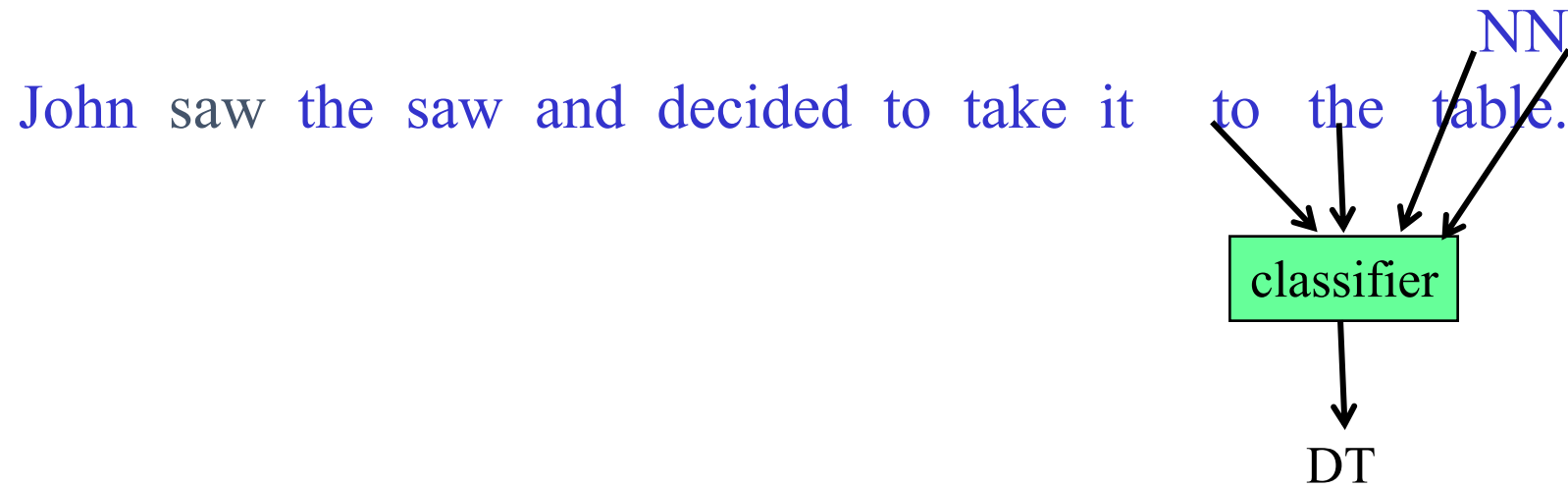


Backward Classification

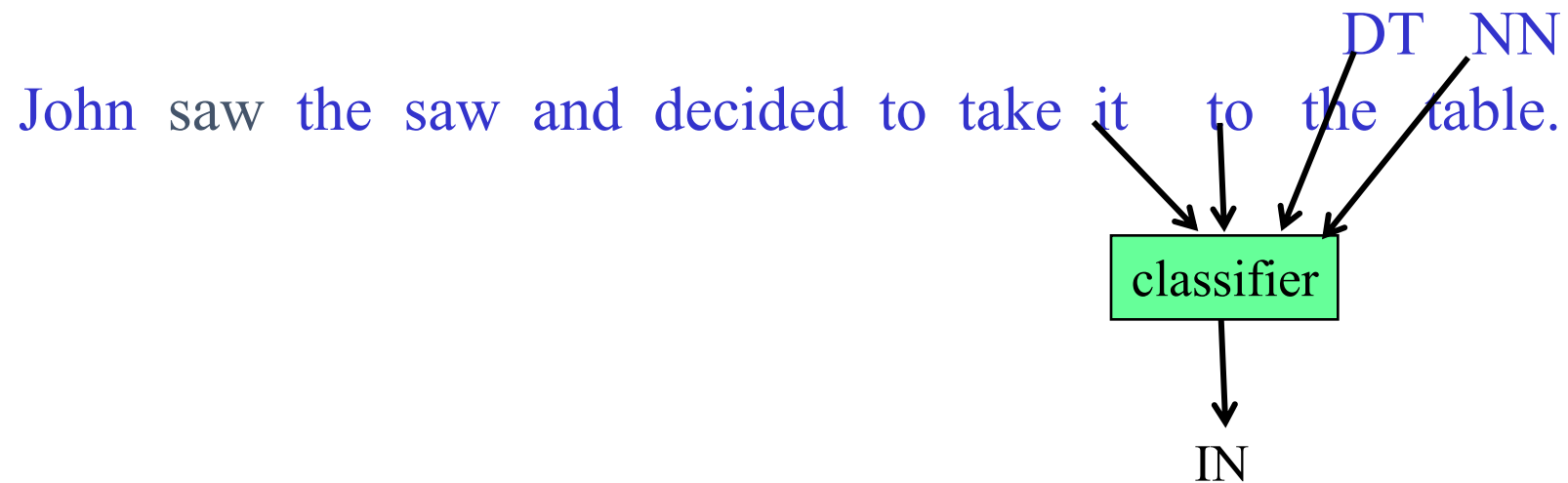
- Disambiguating “to” in this case would be easier backward



Backward Classification



Backward Classification

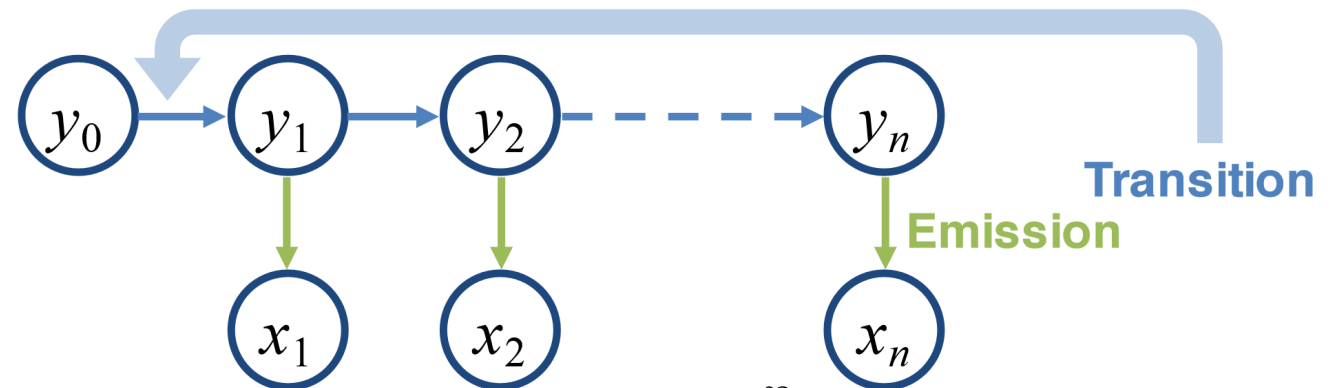


Problems with Sequence Labeling as Classification

- Not easy to integrate information from category of tokens on both sides
- Difficult to propagate uncertainty between decisions and “collectively” determine the most likely joint assignment of categories to all of the tokens in a sequence

Probabilistic Sequence Models

- Probabilistic sequence models allow integrating uncertainty over multiple, interdependent classifications and collectively determine the most likely global assignment
- Classic solution: Hidden Markov Models



$$p(x_1 \dots x_n, y_1 \dots y_n) = q(STOP|y_n) \prod_{i=1}^n q(y_i|y_{i-1})e(x_i|y_i)$$